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Education

Sep 2023 – Nov 2026
Abomey-Calavi, Bénin

Bachelor of Science in Artificial Intelligence

IFRI, University of Abomey-Calavi

Thesis: *Solving the multi-product vehicle routing problem with changeover costs with constraint programming and insertion sequence variables.*

GPA: 16/20

Relevant Coursework: Linear Algebra, Probability & Statistics, Operations Research, Optimization, Algorithms, Graph Theory, Calculus, Machine Learning, Computer Vision, Language Processing

Research Interests

Operations research, production planning and scheduling, Industrial systems; Applied Machine learning; Explainable AI.

Research Experience

Sep 2026 – Sep 2027
Cotonou, Bénin

Research Engineer

[Ai4Innov Technologies](#)

Supervisor: Dr. [Gilles Hacheme](#)

- Fellowship: Working with satellite imagery and geospatial data to build machine learning and deep learning models to support climate resilience and food security work across Benin and the wider region.

Mar 2026 – Aug 2026
Abomey-Calavi, Bénin

Research Intern

[LRSIA, University of Abomey-Calavi](#)

Supervisor: Dr. [Ratheil Houndji](#)

- Formulated the multi-product vehicle routing problem with changeover costs ([MPVRP-CC](#)) as a mixed-integer linear program and implemented exact experiments in Gurobi to establish baseline solutions.
- Conducted research on [sequence variables](#) in constraint programming (CP) for routing problems.
- Proposed a hybrid CP/construction-heuristic approach for the MPVRP-CC using sequence variables, and evaluated it against the MILP baseline.
- Supervised the second-year AI class project (Machine Learning course) and contributed to the [ifri-mini-ml-lib](#) pedagogical library.
- Contributed to the lab's internal projects and to technology monitoring.
- **Tools:** Gurobi, MaxiCP, MiniCP, Python, Java

Work Experience

Jul 2026
Cotonou, Bénin

Applied Machine Learning Mentor

[Sèmè City](#)

- Mentored the top 20 participants of the National Olympiad of Artificial Intelligence (NOAI) 2026 in machine learning and technical problem-solving.
- At the International Olympiad in Artificial Intelligence (IOAI), two mentees earned a bronze medal and an honorable mention, respectively, and the team ranked third in Africa. [\[Link\]](#)

Jun 2025 – Aug 2025
Johannesburg, South Africa

Machine Learning Engineer Intern

[eTihuku](#)

Supervisor: Dr. [Jeremiah Olamijuwon](#)

- Developed a multi-modal NLP engine analyzing customer feedback from social media and phone calls to help insurance companies identify churn risk signals.
- Implemented sentiment analysis with fine-tuned DistilBERT models and emotion recognition through zero-shot classification.
- Applied BERTopic for topic modeling to uncover recurring customer pain points and themes.
- Integrated the Google Translate API to support multilingual reviews.
- **Tools:** PyTorch, Hugging Face Transformers, BERTopic, Airflow

Selected Projects

- **waste-sorting robot:** Smart waste-sorting robot combining real-time object detection (YOLOv5, TensorRT) with a ROS2-controlled robotic arm. [\[Code Repository\]](#)
- **opti'plan:** Co-built an automated scheduling model that reduced the university's thesis defense planning time from manual process of 3 days to 5 minutes using Constraint programming.
- **ifri-mini-ml-lib:** Co-built an open-source Python library that reimplements classic AI algorithms from scratch for educational purposes; published on PyPI and used by IFRI students. [\[Code Repository\]](#)
- **automated real estate mapping:** Built an ETL framework to link address data, cadastral information, companies, and executives, in order to automatically generate enriched customers datasets for prospecting and analysis use cases. [\[Code Repository\]](#)

Conferences

Aug 2026 Lagos, Nigeria	Deep Learning Indaba Presented, <i>A MILP Framework for Multi-Product Vehicle Routing Problem with Changeover Costs</i> [Poster]
Aug 2025 Abomey-Calavi, Bénin	ACP Summer School <i>Constraint Programming for Sustainable Development</i> [Certificate]

Presentations

Nov 2025 Abomey-Calavi, Bénin	Benin Workshop on AI <i>Evolutionary Algorithms: Principles, Variants, Application</i> [Material]
Aug 2025 Ouidah, Bénin	Young Girls' Leadership Camp <i>Young Woman, Take the Lead in Shaping Your Future with AI</i> [Link] Co-presented with Marie-Paule Lima
Oct 2024 Cotonou, Bénin	Future of AI Community <i>Introduction to GPU Programming for AI</i> [Material] Co-presented with Auster Hanou

Honours and Awards

2023 - 2026	Beninese Government Scholarship , 2100\$ awarded for academic excellence in the Baccalaureate .
2020 - 2026	Vallet Foundation Scholarship , 4900\$ awarded for academic excellence in the BEPC .
2025	2nd Place , Tekbot Robotics Challenge 2025. [Link]
2024	1st Prize , IndabaX Benin Republic Hackathon 2024. [Certificate]

Memberships

Feb 2025 - Present	FRIARE Africa R&D contributor working on AI systems ethics and bias evaluation in datasets and models. [Link]
Jun 2025 - Aug 2026	THE LAB Co-led a student AI awareness community in Bénin. [Link]
Jan 2025 - Jul 2026	IFRI AI & IoT Club Co-led the AI & IoT student club: delivered introductory sessions on Machine Learning and Optimization Algorithms, coordinated technical activities, and co-supervised AI projects. [GitHub]
Mar 2026 - Apr 2026	Applied Machine Learning Days (AMLD) 2026 Co-organized the first edition of the Applied Machine Learning Days in Benin Republic. [Certificate]

Skills

Programming	Python, Java, C++, Bash
Machine Learning	PyTorch, JAX, Scikit-learn, Transformers
Optimization	Linear and Constraint Programming, Metaheuristics; Gurobi, MaxiCP
Tools	Git, Linux, Docker, FastAPI, GitHub Actions
Languages	Fongbé , French (native), English (B2)

Other Experience

2018 – Present
Bénin

Family Business Operations *(Part-time)*
Destins Croisés B.

- Support the day-to-day operations of my family’s restaurant and guesthouse, including customer service, reservations, purchasing, inventory, cash handling, and tracking daily income and expenses.

References

1. Dr. Ratheil Houndji, Associate Professor in Artificial Intelligence, University of Abomey-Calavi, Bénin
2. Méléne Tonou, PhD Student in Artificial Intelligence, University of Abomey-Calavi, Bénin
3. Dr. Gilles Hacheme, Senior Research Scientist, Microsoft AI for Good Lab, Kenya

***Contact details available upon request.